

DATA CLASSIFICATION & DATA LOSS PREVENTION IMPLEMENTATION

Microsoft Purview - Sensitivity Labels, Auto-Labeling & DLP Policies

M365 Security Portfolio - Project 3

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Date	September 2026
Platform	Microsoft Purview (M365 E5 Trial Environment)
Framework Alignment	ISO 27001:2022, SOC 2, PCI DSS, HIPAA
Prerequisites	Project 1 - IAM Framework (Entra ID), Project 2 - Endpoint Compliance (Intune)
GitHub	github.com/ethicalkaps

Executive Summary

This document presents a complete Data Classification and Data Loss Prevention (DLP) program implemented in Microsoft Purview within a Microsoft 365 E5 environment. The project demonstrates hands-on capability in designing a four-tier sensitivity label taxonomy, configuring content marking (headers, footers, watermarks), building auto-labeling rules for automated sensitive data detection, and creating DLP policies that actively prevent unauthorized data disclosure.

Data classification and DLP are foundational requirements across every major compliance framework. ISO 27001 requires information classification and handling (A.8.2). PCI DSS requires protection of cardholder data (Requirements 3 and 4). HIPAA requires safeguarding of Protected Health Information. SOC 2's Confidentiality criteria requires that information designated as confidential is protected as committed. This project operationalizes all of these requirements through Microsoft Purview's integrated classification and protection platform.

This project connects directly to Projects 1 and 2. The sensitivity labels defined here are referenced by Conditional Access policies in Entra ID and compliance policies in Intune, creating a unified data protection chain from identity through endpoint to content - a core principle of zero-trust data security.

All configurations were performed in a personal M365 E5 trial environment for portfolio demonstration purposes. No production systems or employer data were used.

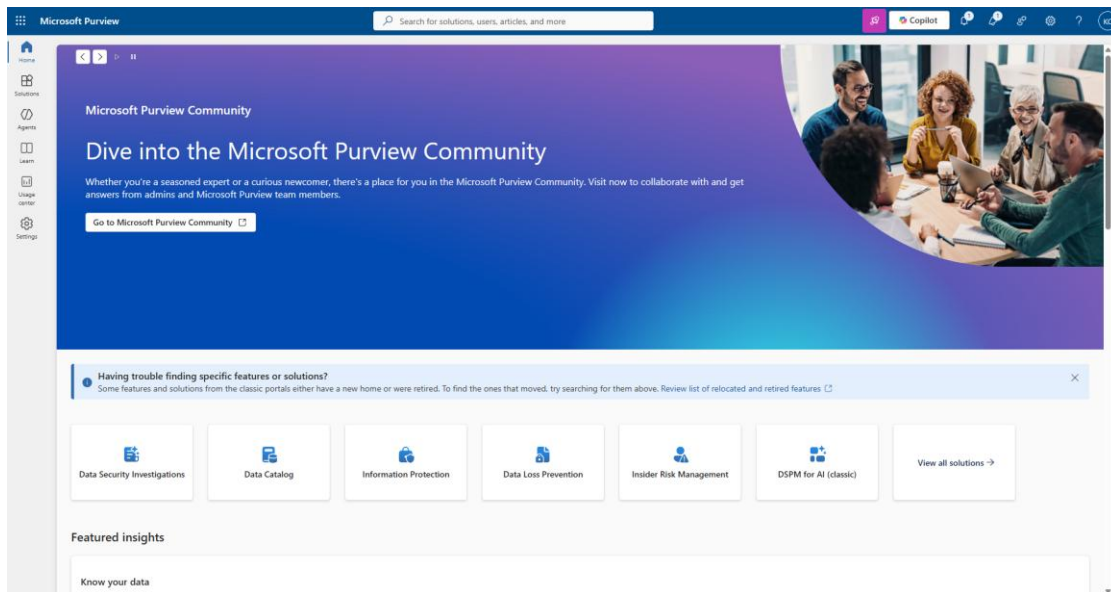
Compliance Mapping Overview

Configuration	ISO 27001 Control	SOC 2 / PCI DSS / HIPAA
4-tier sensitivity labels	A.8.2 - Information classification	SOC 2 Confidentiality / PCI DSS Req 3
Content marking (headers/footers)	A.8.2.2 - Labelling of information	SOC 2 CC6.1 / HIPAA - PHI marking
Watermarks on Highly Confidential	A.8.2.2 - Labelling of information	PCI DSS Req 9
Encryption on Confidential content	A.8.24 - Use of cryptography	PCI DSS Req 3, 4 / SOC 2 CC6.1
Auto-label: credit card detection	A.8.2 - Classification	PCI DSS Req 3 - Identify cardholder data
Auto-label: PII detection	A.8.2 - Classification	HIPAA - PHI protection / SOC 2 Privacy
DLP-001: Block external sharing	A.8.12 - Data leakage prevention	SOC 2 CC6.1 / PCI DSS Req 3, 7
DLP-002: Payment card protection	A.8.12 - Data leakage prevention	PCI DSS Req 3, 4 / SOC 2 CC6.1
Label publishing policy	A.8.2 - Classification	SOC 2 CC6.1 - Organizational controls
Audit logging for labeling	A.12.4.1 - Event logging	SOC 2 CC7.2 - Monitoring

1. Environment Overview

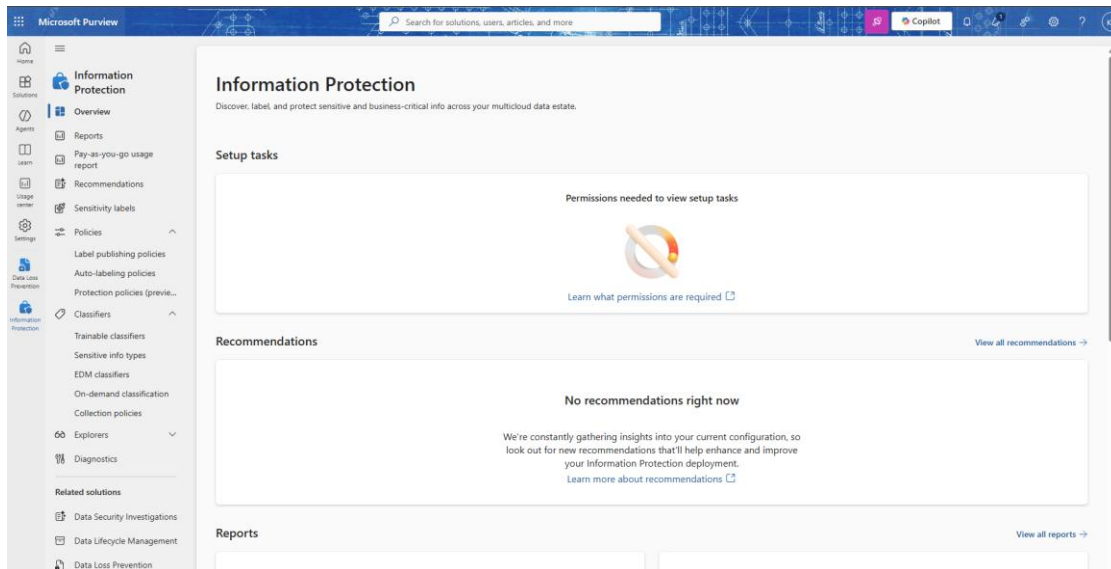
Microsoft Purview is the unified data governance, risk, and compliance platform within the Microsoft 365 ecosystem. It provides Information Protection (sensitivity labels, encryption, content marking), Data Loss Prevention (policy-based enforcement preventing unauthorized data sharing), and Data Classification (automated discovery and categorization of sensitive data across the organization).

1.1 Purview Home Dashboard



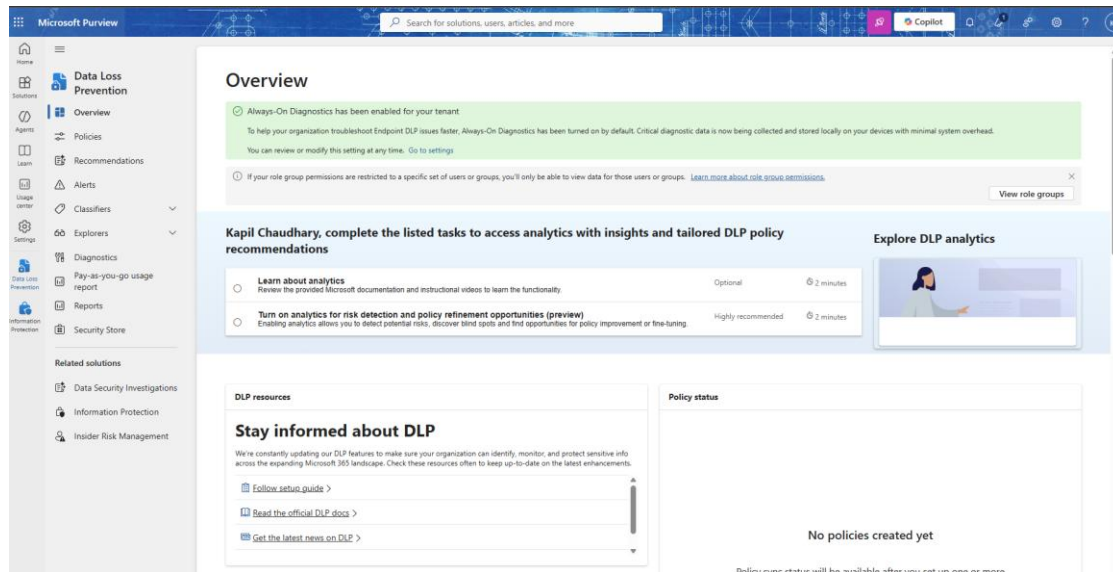
1.2 Information Protection Overview

The Information Protection module is where sensitivity labels are created, published, and managed. This is the classification engine that defines what each sensitivity tier means and what protections apply.



1.3 Data Loss Prevention Overview

The DLP module is where enforcement policies are created. While classification labels define sensitivity, DLP policies actively prevent sensitive data from leaving the organization through email, file sharing, or collaboration tools.



2. Sensitivity Labels - Four-Tier Classification Taxonomy

A four-tier sensitivity label taxonomy was implemented following the industry-standard classification scheme used by most enterprise organizations: Public, Internal, Confidential, and Highly Confidential. Each tier applies progressively stricter protections - from no restrictions on Public content through full encryption, watermarking, and sharing restrictions on Highly Confidential content.

The existing Microsoft default labels (Confidential, Highly Confidential with sub-labels) were reviewed and customized with content marking. Additional labels (Public, Internal) were created to complete the four-tier taxonomy. This approach mirrors real-world practice - inheriting and improving existing label structures rather than rebuilding from scratch.

2.1 Label Creation Process

Each label was configured with scope (Files, Emails), content marking settings, and appropriate protection levels. The creation wizard captures label name, description for users, scope, and protection settings.

Edit sensitivity label

- Label details
- Scope
- Items
- Groups & sites
- Finish

Provide basic details for this label

The protection settings you choose for this label will be immediately enforced on the items or content containers to which it's applied. Labeled files will be protected wherever they go, whether they're saved in the cloud or downloaded to a computer.

Name *

default17b-0d19-0005-0001-bu88714345d2

Display name *

Public

Label priority

1

Description for users *

Information approved for public disclosure. No restrictions on sharing.

Description for admins

Enter a description that's helpful for admins who will manage this label

Label color



Next

Cancel

Edit sensitivity label

- Label details
- Scope
- Items
- Groups & sites
- Finish

Define the scope for this label

Labels can be applied to data assets and containers (like SharePoint sites and Teams). Let us know where you want this label to be used so we can show you the related protection settings. [Learn more about label scopes](#)

☒ Files & other data assets

Label files and data assets in Microsoft 365, Fabric (includes Power BI), Viva Engage, and Azure.

☒ Emails

Label messages sent from all versions of Outlook.

☒ Meetings

Label calendar events and meetings schedules in Outlook and Teams.

☐ Groups & sites

Configure privacy, access control, and other settings to protect labeled Teams, Microsoft 365 Groups, Entra Security Groups, SharePoint sites, Viva Engage content, and Loop workspaces.

To apply sensitivity labels to Teams, SharePoint sites, and Microsoft 365 Groups, you must first [complete these steps](#) to enable the feature.

Wondering where the **schematized data assets** option went? To apply this label to data assets in Azure, select "Files and other data assets" above and then add this label to an auto-labeling policy that's scoped to Azure storage and Azure SQL. [Learn more](#)



Edit sensitivity label

- Label details
- Scope
- Items
- Groups & sites
- Finish

Choose protection settings for the types of items you selected

The protection settings you configure will be enforced when the label is applied to items in Microsoft 365.

☐ Control access

Control who can access and view labeled items.

☐ Apply content marking

Add custom headers, footers, and watermarks to labeled items.



Edit sensitivity label

- Label details
- Scope
- Items
- Auto-labeling for files and emails
- Groups & sites
- Finish

Auto-labeling for files and emails

When users edit Office files or compose, reply to, or forward emails from Outlook that contain content matching the conditions you choose here, we'll automatically apply this label or recommend that they apply it themselves. [Learn more about auto-labeling for Microsoft Purview](#)

To automatically apply this label to files that are already saved in SharePoint and OneDrive or emails that are already processed by Exchange, you must create an auto-labeling policy. [Learn more about auto-labeling policies](#)

Auto-labeling for files and emails



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Edit sensitivity label

Label details

Scope

Items

Groups & sites

Finish

Define protection settings for groups and sites

Protection settings

These settings apply to teams, groups, and sites that have this label applied. They don't apply directly to the files stored in those containers.

[Learn more about these settings](#)

Privacy and external user access

Control the level of access that internal and external users will have to labeled teams and Microsoft 365 Groups.

External sharing and Conditional Access

Control external sharing and configure Conditional Access settings to protect labeled SharePoint sites.

Private teams discoverability and shared channel settings

Decide whether private teams will be discoverable in searches and control the types of teams that can be invited to shared channels.

Auto apply settings

Apply a label to channel meetings

For Teams channels that have this label applied, you can also select a label to automatically apply to any meetings created from the channel. Users won't be able to change or remove the label. [Learn more about this setting](#)

Public

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Edit sensitivity label

Label details

Scope

Items

Groups & sites

Finish

Review your settings and finish

Name

defa4170-0d19-0005-0001-bc88714345d2

Display name

Public

[Edit](#)

Description for users

Information approved for public disclosure. No restrictions on sharing.

[Edit](#)

Scope

Files & other data assets, Email, Meetings, Site, UnifiedGroup

[Edit](#)

Access control

None

[Edit](#)

Content marking

None

[Edit](#)

Auto-labeling for files and emails

None

[Edit](#)

Group settings

None

[Edit](#)

Site settings

None

[Edit](#)

Meetings settings

[Edit](#)

Back

Save label

2.2 Label 1 - Public

Information approved for public disclosure. No restrictions on sharing, no content marking, no encryption. This is the default label applied to all new documents - ensuring every document has a classification from creation, with users upgrading the label when content becomes sensitive.

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Information Protection

Sensitivity labels

+ Create Publish labels Export Refresh 5 items

	Name	Priority	Scope	Created by	Last modified
☐	Personal	0	Files & other data assets, Email, Me...	Microsoft Corporation	Sep 14, 2026 3:02:57 PM
☐	Public	1	Files & other data assets, Email, Me...	Microsoft Corporation	Sep 19, 2026 9:02:51 PM
☐	General	2		Microsoft Corporation	Sep 19, 2026 3:03:08 PM
☐	Confidential	3		Microsoft Corporation	Sep 14, 2026 3:03:18 PM
☐	Highly Confidential	9		Microsoft Corporation	Sep 14, 2026 3:03:39 PM

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Create policy

Labels to publish

Admin units

Users and groups

Settings

Name

Finish

Choose sensitivity labels to publish

When published, the labels you choose here will be available in specified users' Office apps (Word, Excel, PowerPoint, Outlook, Teams), SharePoint and Teams sites, and Microsoft 365 Groups.

Sensitivity labels to publish

Choose sensitivity labels to publish

Sensitivity labels to publish

If you select a sublabel, the corresponding parent will also be published automatically.

Search for specific labels

1 selected

Label	Scope
☐ Personal	Files & other data assets, Email, Meetings
☒ Public	Files & other data assets, Email, Meetings
☐ General/Anyone (unrestricted)	Files & other data assets, Email, Meetings
☐ General/All Employees (unrestricted)	Files & other data assets, Email, Meetings
☐ Confidential/Anyone (unrestricted)	Files & other data assets, Email, Meetings
☐ Confidential/All Employees	Files & other data assets, Email, Meetings
☐ Confidential/Trusted People	Files & other data assets, Email, Meetings
☐ Highly Confidential/All Employees	Files & other data assets, Email, Meetings
☐ Highly Confidential/Specified People	Files & other data assets, Email, Meetings

Next Add Cancel

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Create policy

Labels to publish

Admin units

Users and groups

Settings

Name

Finish

Assign admin units

Choose the admin units you'd like to assign this policy to. Admin units are created in Microsoft Entra ID and restrict the policy to a specific set of users or groups. Your selections will affect the location options available to you in the next step. If you want to assign this policy to all users and groups, select 'Next' and proceed. [Learn more about admin units](#)

+ Add or remove admin units

Admin units

Full directory

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Create policy

Labels to publish

Admin units

Users and groups

Settings

Name

Finish

Publish to users and groups

The labels you selected will be available for the users, distribution groups, mail-enabled security groups, and Microsoft 365 Groups you choose here.

If your role group permissions are restricted to a specific set of users and groups, you'll only be able to publish labels for those users and groups. [Learn more about role group permissions](#)

View role groups

Location	Scope	Actions
☒ Exchange email	All users & groups	Edit

2.3 Label Publishing - Standard Label Policy

Labels were published through a label policy (LP-001: Standard Label Policy) assigned to all users. The policy sets Public as the default label, requires justification when downgrading a label (e.g., changing from Confidential to Public), and defines the organizational scope of label availability.

This screenshot shows the 'Create policy' wizard for documents. The left sidebar lists the steps: Labels to publish, Admin units, Users and groups, Settings (selected), Documents, Emails, Meetings, Sites and Groups, Engage, Fabric and Power BI, Name, and Finish. The main content area is titled 'Default settings for documents' and 'Apply a default label to documents'. It explains that the chosen label will be applied to Word, Excel, and PowerPoint documents. A dropdown menu for 'Default label' shows 'Public' selected, with 'None' and 'Public' as visible options. The 'Public' option is highlighted with a red box. At the bottom, 'Back' and 'Next' buttons are visible, with 'Next' highlighted by a red box.

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Create policy

- Labels to publish
- Admin units
- Users and groups
- Settings**
- Documents
- Emails
- Meetings
- Sites and Groups
- Engage
- Fabric and Power BI
- Name
- Finish

Default settings for documents

Apply a default label to documents

The label you choose will automatically be applied to Word, Excel, and PowerPoint documents when they're created or modified. Users can always select a different label to better match the sensitivity of their document. [Learn which Office app versions support this setting](#)

Default label

Public

None

Public

Back Next

This screenshot shows the 'Create policy' wizard for emails. The left sidebar lists the steps: Labels to publish, Admin units, Users and groups, Settings (selected), Documents, Emails (selected), Meetings, Sites and Groups, Engage, Fabric and Power BI, Name, and Finish. The main content area is titled 'Default settings for emails' and 'Apply a default label to emails'. It explains that the chosen label will be applied to new and existing, unlabeled emails. A dropdown menu for 'Default label' shows 'Public' selected. Below this, there is a checkbox for 'Require users to apply a label to their emails' which is unchecked. The next section is 'Inherit label from attachments', which explains that if an email is labeled and attachments with higher priority labels are added, the email's label can be replaced. There is a checkbox for 'Email inherits highest priority label from attachments' which is unchecked. At the bottom, 'Back' and 'Next' buttons are visible, with 'Next' highlighted by a red box.

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Create policy

- Labels to publish
- Admin units
- Users and groups
- Settings**
- Documents
- Emails**
- Meetings
- Sites and Groups
- Engage
- Fabric and Power BI
- Name
- Finish

Default settings for emails

Apply a default label to emails

The label you choose will automatically be applied to new and existing, unlabeled emails. Users can always change the default label before they send the message. [Learn which Outlook versions support these settings](#)

Default label

Public

☐ Require users to apply a label to their emails ⓘ

Inherit label from attachments

If an email is labeled and attachments with higher priority labels are added, you can automatically replace the email's label with the highest priority attachment label or recommend that users do it themselves. If the email isn't labeled, it inherits the highest priority label from attachments. [Learn more about label inheritance](#)

☐ Email inherits highest priority label from attachments

Back Next

Create policy

- Labels to publish
- Admin units
- Users and groups
- Settings
- Name**
- Finish

Name your policy

Name *
LP-001: Standard Label Policy

Enter a description for your sensitivity label policy
Every new document starts as Public unless changed.

Back

Next

Create policy

- Labels to publish
- Admin units
- Users and groups
- Settings
- Name
- Finish**

Review and finish

Name
LP-001: Standard Label Policy
[Edit](#)

Description
Every new document starts as Public unless changed.
[Edit](#)

Publish these labels
Public
[Edit](#)

Publish to users and groups
Exchange email - 1 account
[Edit](#)

Policy settings
Default label for documents is: Public
Default label for emails is: Public
Default label for meetings is: Public
[Edit](#)

Back

Submit

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Create policy

Labels to publish

Admin units

Users and groups

Settings

Name

Finish

✓ New policy created

It can take up to 24 hours to publish the labels to the selected users' apps.

Next steps
[Review data classification reports](#) to see how labels are being used
[Read guidance](#) on how to educate users about sensitivity labels

2.4 Label 2 - Internal

Business information intended for internal use only. Content marking enabled - header displaying 'INTERNAL - Do Not Distribute Externally' and footer with classification designation. No encryption applied, allowing free internal sharing while clearly marking documents as internal-only.

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New sensitivity label

Label details

Scope

Items

Groups & sites

Finish

Provide basic details for this label

The protection settings you choose for this label will be immediately enforced on the items or content containers to which it's applied. Labeled files will be protected wherever they go, whether they're saved in the cloud or downloaded to a computer.

Name *

Internal

Display name *

Internal

Label priority

By default, this label will be assigned the highest priority, but you can change this after it's created.

Highest

Description for users *

Business information intended for internal use only. Do not share externally without approval.

Description for admins

Enter a description that's helpful for admins who will manage this label.

Label color

Next

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New sensitivity label

Label details

Scope

Items

Groups & sites

Finish

Choose protection settings for the types of items you selected

The protection settings you configure will be enforced when the label is applied to items in Microsoft 365.

Control access

Control who can access and view labeled items.

Apply content marking

Add custom headers, footers, and watermarks to labeled items.

Protect Teams meetings and chats

Configure protection settings for Teams meetings and chats.

To protect Teams meetings and chats, your org must have a Teams Premium license. [Learn more about Teams Premium](#)

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New sensitivity label

Label details

Scope

Items

Content marking

Auto-labeling for files and emails

Groups & sites

Finish

Content marking

Add custom headers, footers, and watermarks to content that has this label applied. [Learn more](#)

☐ All content marking will be applied to documents but only the header and footer will be applied to email messages. If header and footer will also be applied to meeting invites.

Content marking

☒ Add a header [Customize text](#)

☒ Add a footer [Customize text](#)

Customize header text

This text will appear as a header on labeled email messages and documents.

Header text *

Font size

Font color

Align text

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New sensitivity label

Label details

Scope

Items

Content marking

Auto-labeling for files and emails

Groups & sites

Finish

Content marking

Add custom headers, footers, and watermarks to content that has this label applied. [Learn more](#)

☐ All content marking will be applied to documents but only the header and footer will be applied to email messages. If header and footer will also be applied to meeting invites.

Content marking

☐ Add a watermark [Customize text](#)

☒ Add a header [Customize text](#)

☒ Add a footer [Customize text](#)

INTERNAL — Do Not Distribute Externally

Customize footer text

This text will appear as a footer on labeled email messages and documents.

Footer text *

Font size

Font color

Align text

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New sensitivity label

Label details

Scope

Items

Groups & sites

Finish

Review your settings and finish

Name

Internal

Edit

Display name

Internal

Edit

Description for users

Business information intended for internal use only. Do not share externally without approval.

Edit

Scope

Files & other data assets, Email

Edit

Access control

None

Edit

Content marking

Header: INTERNAL — Do Not Distribute Externally

Footer: Classification: Internal — [Company Name]

Edit

Auto-labeling for files and emails

None

Edit

Back

Create label

2.5 Labels 3 & 4 - Confidential and Highly Confidential

The existing Microsoft default Confidential and Highly Confidential parent labels were reviewed and customized. Both include sub-labels providing granular sharing controls: 'All Employees' (accessible to all authenticated organization members), 'Trusted People' (Confidential tier - specific external parties), and 'Specified People' (Highly Confidential tier - named individuals only). Content marking was added including headers, footers, and watermarks for the Highly Confidential tier. This sub-label structure provides more granular access control within each classification tier than flat labels - a design pattern used in mature enterprise environments.

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Information Protection

Overview

Reports

Pay-as-you-go usage report

Recommendations

Sensitivity labels

Policies

Label publishing policies

Auto-labeling policies

Protection policies (previous)

Classifiers

Explorers

Diagnostics

Related solutions

Data Security Investigations

Data Lifecycle Management

Data Loss Prevention

Sensitivity labels

You can now create sensitivity labels with privacy and access control settings for Teams, SharePoint sites, and Microsoft 365 Groups. To do this, you must first [complete these steps](#) to enable the feature.

Sensitivity labels are used to classify email messages, documents, sites, and more. When a label is applied (automatically or by the user), the content or site is protected based on the settings you choose. For example, you can create labels that encrypt files, add content marking, and control user access to specific sites. [Learn more about sensitivity labels](#)

+ Create

Publish labels

Export

Refresh

6 items

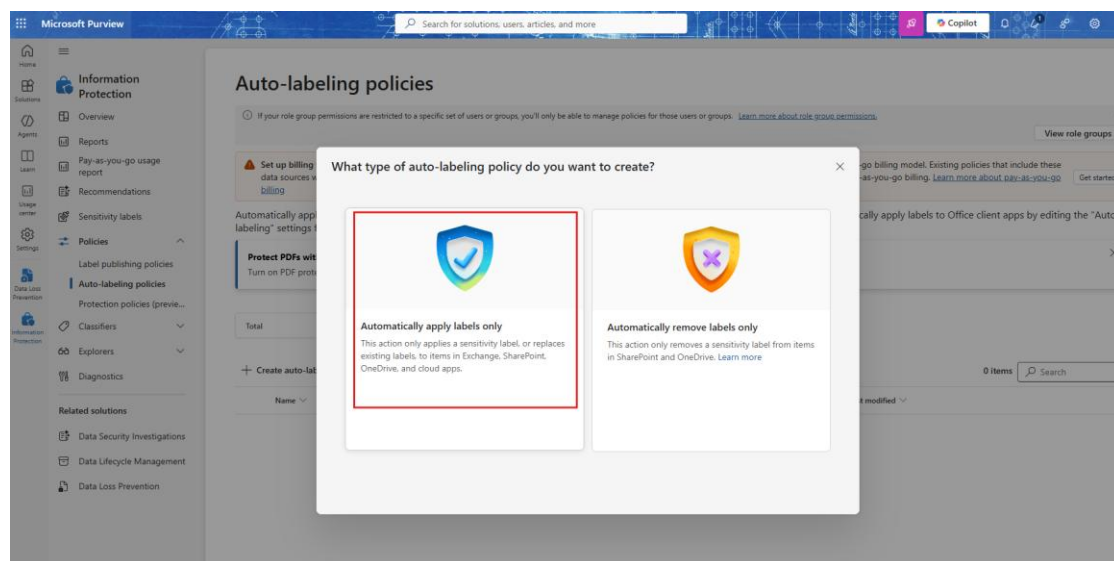
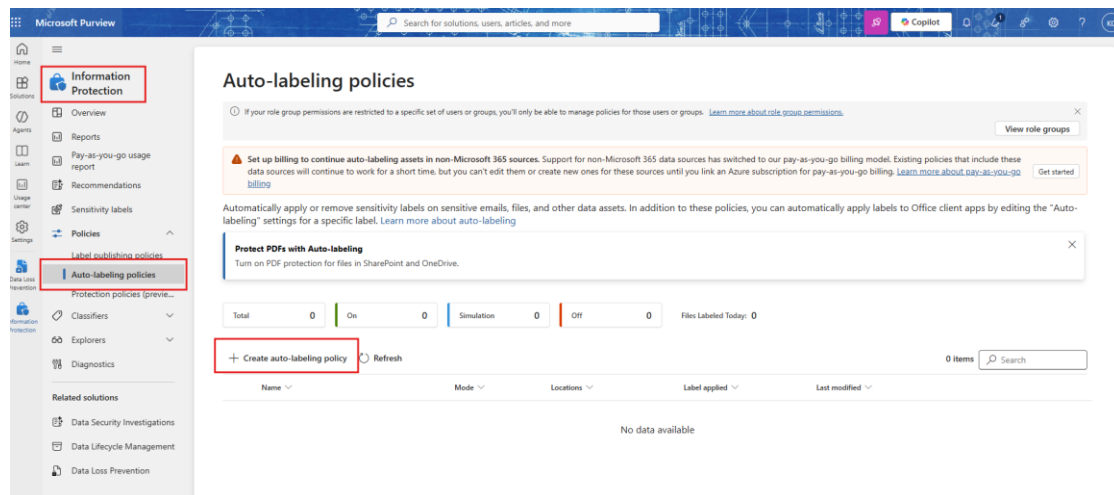
☐	Name	Priority	Scope	Created by	Last modified
☐	Personal	0	Files & other data assets, Email, Mee...	Microsoft Corporation	Sep 14, 2026 3:02:57 PM
☐	Public	1	Files & other data assets, Email, Mee...	Microsoft Corporation	Sep 19, 2026 9:02:51 PM
☐	General	2		Microsoft Corporation	Sep 14, 2026 3:03:06 PM
☐	Confidential	3		Microsoft Corporation	Sep 19, 2026 9:18:42 PM
☐	Anyone (unrestricted)	6	Files & other data assets, Email, Mee...	Microsoft Corporation	Sep 19, 2026 9:23:17 PM
☐	All Employees	7	Files & other data assets, Email, Mee...	Microsoft Corporation	Sep 19, 2026 9:24:10 PM
☐	Trusted People	8	Files & other data assets, Email, Mee...	Microsoft Corporation	Sep 19, 2026 9:25:18 PM
☐	Highly Confidential	9		Microsoft Corporation	Sep 14, 2026 3:03:39 PM
☐	All Employees	10	Files & other data assets, Email, Mee...	Microsoft Corporation	Sep 19, 2026 9:26:29 PM
☐	Specified People	11	Files & other data assets, Email, Mee...	Microsoft Corporation	Sep 19, 2026 9:27:27 PM
☐	Internal	12	Files & other data assets, Email	Kapil Chaudhary	Sep 19, 2026 9:15:06 PM

3. Auto-Labeling - Automated Sensitive Data Detection

Auto-labeling policies automatically detect and classify documents and emails containing sensitive data patterns - credit card numbers, personal identification numbers, health records - without relying on users to label content manually. This is critical for compliance because manual labeling alone is insufficient; users forget, make mistakes, or don't recognize sensitive content. Auto-labeling provides the automated safety net.

3.1 Auto-Labeling Configuration

Auto-labeling policies use Microsoft's built-in sensitive information type detectors - pre-built pattern matching for hundreds of data types including credit card numbers, Social Insurance Numbers, passport numbers, and medical record identifiers.



3.2 AL-001: Auto-Detect Payment Card Data

This policy automatically detects credit card number patterns across Exchange email, SharePoint, and OneDrive. Documents containing credit card data are automatically labeled as Highly Confidential, triggering all associated protections (encryption, content marking, DLP enforcement). Deployed in Simulation mode to evaluate detection accuracy before enforcement.

The screenshot shows the 'Choose info you want this label applied to' step in the Microsoft Purview console. The left sidebar contains a navigation menu with steps: Info to label, Name, Label, Admin units, Locations, Policy rules, Policy mode, and Finish. The main content area has a header 'Choose info you want this label applied to' and a sub-header 'Choose an industry regulation to see the policy templates you can use to classify that info or create a custom policy to start from scratch.' Below this is a warning message: 'Protect content in Microsoft 365 Copilot. You can now prevent Copilot from processing specific types of content (like files with sensitivity labels applied). To try it out, make sure you start with a custom policy below. Learn more about protecting content in Copilot.' There are two input fields: 'Search templates' with a search icon and 'Select location' with a dropdown menu showing 'All countries or regions'. Below these are two columns of results. The 'Categories' column lists: Enhanced, Financial (highlighted with a red box), Medical and health, Privacy, and Custom. The 'Regulations' column lists: Australia Financial Data, Canada Financial Data, France Financial Data, Germany Financial Data, Israel Financial Data, Japan Financial Data, PCI Data Security Standard (PCI DSS) (highlighted with a red box), Saudi Arabia - Anti-Cyber Crime Law, and Saudi Arabia Financial Data. To the right of these columns is a detailed description for 'PCI Data Security Standard (PCI DSS)' which states: 'Helps detect the presence of information subject to PCI Data Security Standard (PCI DSS), including information like credit card or debit card numbers.' Below this description is a section 'Protect this information:' with a bullet point 'Credit Card Number'. At the bottom of the main content area is a 'Next' button highlighted with a red box.

The screenshot shows the 'Name your auto-labeling policy' step in the Microsoft Purview console. The left sidebar contains a navigation menu with steps: Info to label, Name, Label, Admin units, Locations, Policy rules, Policy mode, and Finish. The main content area has a header 'Name your auto-labeling policy' and a sub-header 'This policy will automatically apply a label to items that match rules and conditions you'll define.' Below this is a 'Name*' field with a text input containing 'AL-001: Auto-Detect Payment Card Data' (highlighted with a red box). Below the name field is a 'Description' field with a text input containing 'Enter a description for your new auto-labeling policy'. At the bottom right of the description field is a small icon of two people.

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Auto-labeling policies > New policy

Info to label

Name

Label

Admin units

Locations

Policy rules

Policy mode

Finish

Choose a label to auto-apply

Users will see this label applied to files that match the rules and conditions you chose. [Learn more about labels](#)

Label to auto-apply
+ Choose a label

Choose a sensitivity label

Search for specific labels

1 selected

Label	Type
☐ Personal	Files & other data assets, Email, Meetings
☐ Public	Files & other data assets, Email, Meetin...
☐ General/Anyone (unrestricted)	Files & other data assets, Email, Meetings
☐ General/All Employees (unrestricted)	Files & other data assets, Email, Meetings
☐ Confidential/Anyone (unrestricted)	Files & other data assets, Email, Meetings
☐ Confidential/All Employees	Files & other data assets, Email, Meetings
☐ Confidential/Trusted People	Files & other data assets, Email, Meetings
☒ Highly Confidential/All Employees	Files & other data assets, Email, Meetings
☐ Highly Confidential/Specified People	Files & other data assets, Email, Meetings
☐ Internal	Files & other data assets, Email

Back

Next

Add

Cancel

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Auto-labeling policies > New policy

Info to label

Name

Label

Admin units

Locations

Policy rules

Policy mode

Finish

Choose locations where you want to apply the label

Exchange will automatically apply the label to unlabeled emails, regardless of which device or platform is used to send and receive the email. OneDrive and SharePoint will automatically apply the label to unlabeled Office documents and PDFs.

Before you continue creating this Microsoft Purview autolabeling policy, turn off or delete any Microsoft Defender for Cloud Apps file policies for non-Microsoft locations. Running both at once can cause unexpected policy enforcement.

You can now apply labels to assets in non-Microsoft 365 locations like Azure and Amazon S3. Register and scan those data sources from Data Map so that they can be included in auto-labeling policies. [Learn more](#)

If your role group permissions are restricted to a specific set of users or groups, you'll only be able to apply the label to those users or groups. [Learn more about role group permissions](#)

View role groups

Location	Scope	Actions
☒ Exchange email	All users & groups	Edit
☒ SharePoint sites	All sites	Edit
☒ OneDrive accounts	All users & groups	Edit

Back

Next

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Copilot

Auto-labeling policies > New policy

Info to label

Name

Label

Admin units

Locations

Policy rules

Policy mode

Finish

Set up common or advanced rules

Rules are made up of conditions that define what content the label is applied to. Choose common rules to define one set of rules that will apply to all locations you selected or choose advanced rules to define different rules for each location.

Common rules. Define one set of common rules for all locations.

Common rules consist of conditions for specifying what sensitive info to detect and whether to apply the label to items shared inside or outside your organization. If you selected Exchange and want access to more email conditions, choose advanced rules.

Credit Card Number

☐ Advanced rules. Define specific rules for each location.

Back

Next

Auto-labeling policies > New policy

Info to label

Name

Label

Admin units

Locations

Policy rules

Policy mode

Finish

Decide if you want to test out the policy now or later

To help you determine whether the label will be applied to the correct items, you'll need to run the policy in simulation mode before turning it on. You can do this right away or wait until later.

Turn on policy

Run policy in simulation mode

Leave policy turned off

We'll gather items that match the policy but labels won't be applied yet. It's highly recommended that you review these items to decide whether the policy needs to be refined or if it's ready to be turned on.

Automatically turn on policy if not modified after 7 days in simulation.
The 7-day period will restart each time the policy is modified while in simulation.

Your settings will be saved, but the policy will be inactive until you run it in simulation mode and then turn it on.



Auto-labeling policies > New policy

Info to label

Name

Label

Admin units

Locations

Policy rules

Policy mode

Finish

Review and finish

Policy name
AL-001: Auto-Detect Payment Card Data
[Edit](#)

Mode
Simulation

Status
Simulation in progress

Label
Highly Confidential/All Employees
[Edit](#)

Policy template type
PCI Data Security Standard (PCI DSS)
[Edit](#)

Sensitive Info Type
Credit Card Number

Admin units
None

Apply to content in these locations
Exchange email All
SharePoint sites All
OneDrive accounts All
[Edit](#)

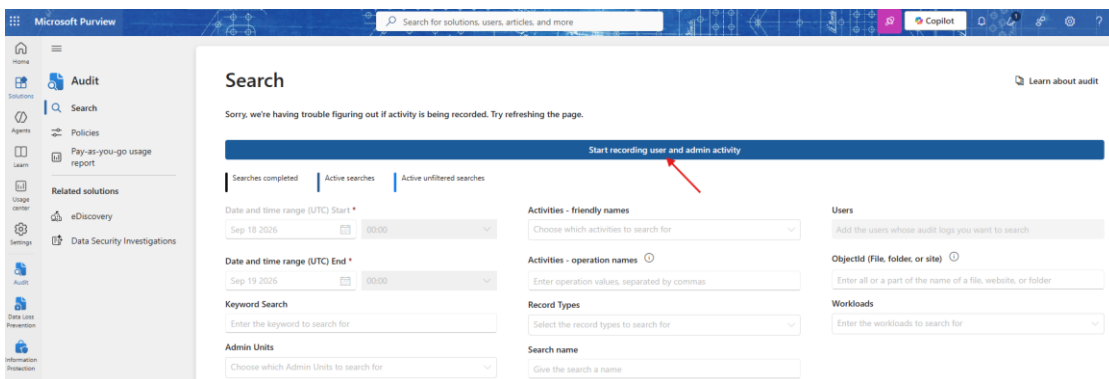
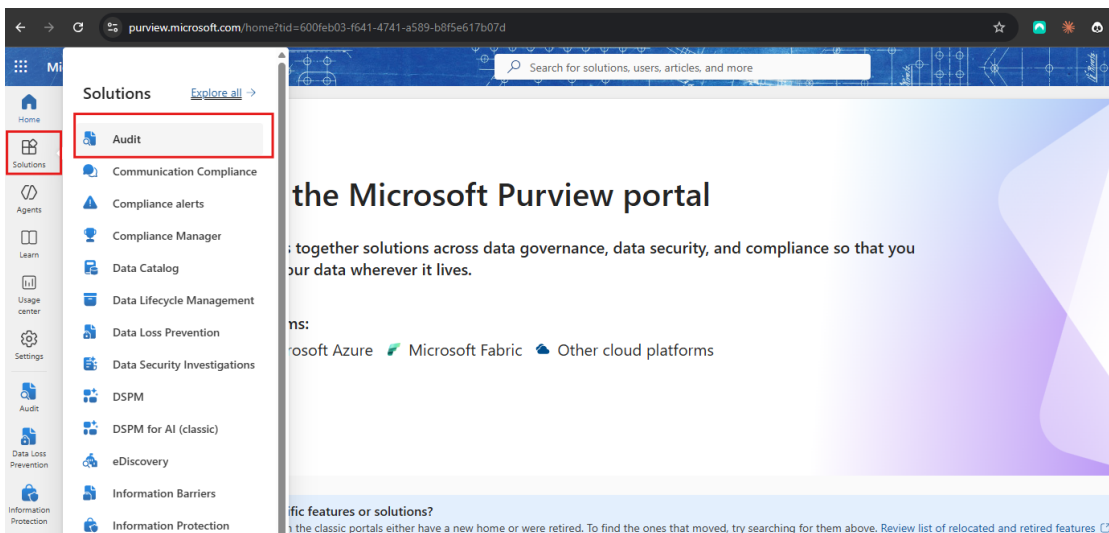
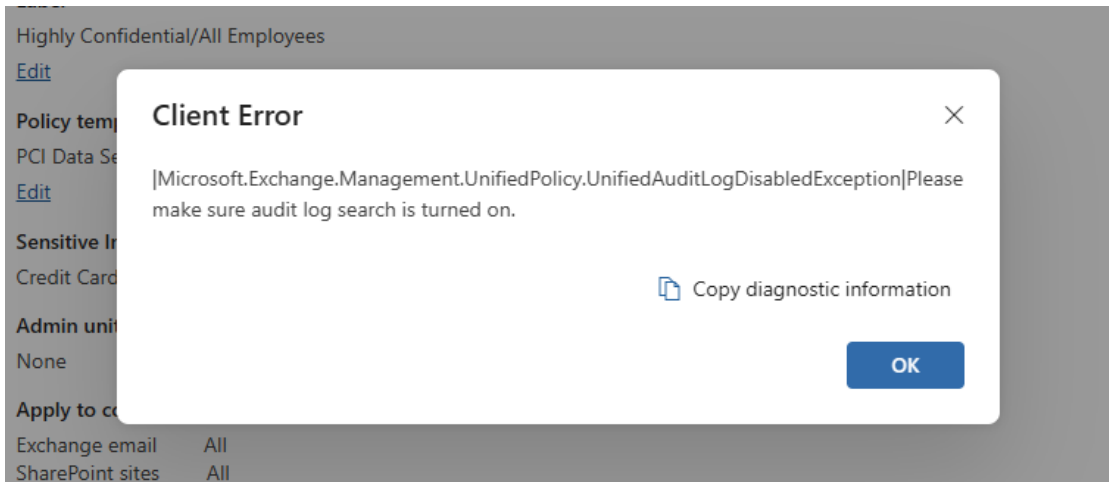
Exclude content from these locations
Exchange email None
SharePoint sites None
OneDrive accounts None
[Edit](#)

Back

Create policy

3.3 Audit Log Prerequisite

Auto-labeling requires unified audit logging to be enabled. During configuration, an error was encountered because audit log search was not yet activated. This was resolved by navigating to Solutions > Audit and enabling activity recording. This is a common prerequisite in new M365 tenants.



ISO 27001: A.12.4.1 - Event logging. SOC 2: CC7.2 - System monitoring. Audit logging is a prerequisite for both auto-labeling and compliance evidence generation.

4. Data Loss Prevention Policies

DLP policies are the enforcement layer of data protection. While sensitivity labels classify data and auto-labeling detects it, DLP policies actively prevent sensitive data from leaving the organization. They monitor data in motion (email, sharing) and apply actions (block, notify, report) based on the content's sensitivity level and sharing context.

DLP policies use a condition-action model. The sharing context condition must be defined first - whether content is being shared externally, with specific domains, or with anyone. Then content conditions (sensitivity labels, sensitive info types) determine what triggers the policy. Actions like blocking or notifying only apply within that defined sharing context.

4.1 DLP Policy Creation Process

Policies

Use data loss prevention (DLP) policies to help identify and protect your organization's sensitive info. For example you can set up policies to help make sure information in email and docs isn't shared with the wrong people. [Learn more about DLP](#)

Set up pay-as-you-go billing. To use some data loss prevention policy features, you must first link an Azure subscription for pay-as-you-go billing. Your org will be charged based on usage. [Learn more about](#) [Get started](#)

Safeguard sensitive data in Microsoft 365 Copilot interactions

Purview now offers granular Copilot controls, allowing admins to block all processing or allow internal knowledge access while blocking web searches when sensitive data is detected. The default DLP policy is currently in simulation mode and can be turned on to restrict Copilot from processing sensitive prompts or performing web searches.

Enforce policy

+ Create policy **Export policies** **Refresh** **Copilot**

Name	Priority	Mode	Policy sync status	Last modified
Default Office 365 DLP policy	0	On	Sync completed	Sep 14, 2026 3:02 PM
Default DLP policy - Protect sensitive M365 Copilot interactions	1	Test with notifications	Sync completed	Sep 14, 2026 3:02 PM
Default policy for Teams	2	On	Sync completed	Sep 14, 2026 3:02 PM

What info do you want to protect?

Let us know what type of data you want your policy to cover, and we'll guide you through a tailored setup experience. [Learn more about these options](#)

Enterprise applications & devices

Includes protection for data across all connected sources in your organization — including Microsoft 365 locations, Microsoft Copilot experiences, data connectors, enterprise-registered apps, and managed apps accessed with the Edge for Business browser.

Inline web traffic

Includes protection for data transferred in real-time with unmanaged cloud apps through the Edge for Business browser and network SASE/SSSE integrations.

Start with a template or create a custom policy

Choose an industry regulation to see the DLP policy templates you can use to protect that info or create a custom policy start from scratch. If you need to protect labeled content, you'll be able to choose labels later. [Learn more about DLP policy templates](#)

Search templates

Select location

All countries or regions

Categories

- Enhanced
- Financial
- Medical and health
- Privacy
- Custom**

Regulations

- Custom policy**

Custom policy

Create a custom policy from scratch. You will choose the type of content to protect and how you want to protect it.

Microsoft Purview

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Policy mode

Finish

Name your DLP policy

Create a DLP policy to detect sensitive data across locations and apply protection actions when the conditions match.

Name *

DLP-001: Block External Sharing of Confidential Content

Description

Enter a friendly description for your policy

Microsoft Purview

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Policy mode

Finish

Choose where to apply the policy

We'll apply the policy to data that's stored in the locations you choose.

- The Instances policy location will be retired in early 2027 as part of the Microsoft Defender for Cloud Apps file policy retirement. For non-Microsoft connected apps, create new DLP policies in Purview using a specific location (for example, Box or Google Workspace) instead.
- Before you continue creating this Microsoft Purview DLP policy, turn off or delete any Microsoft Defender for Cloud Apps file policies for non-Microsoft locations. Running both at once can cause unexpected policy enforcement.
- Conditional Access policy not found. Conditional access policies are required for managed app policies. [Learn more](#). Check your conditional access policy settings or create a new one in Entra ID [here](#).
- Edge for business protection settings not found. This is required for managed app policies. [Learn more](#). Check your Edge for business settings [here](#).
- If your role group permissions are restricted to a specific set of users or groups, you'll only be able to apply this policy to those users or groups. [Learn more about role group permissions](#).
- Protecting sensitive info in on-premises repositories (SharePoint sites and file shares) is now in preview. Note that there are prerequisite steps needed to support this new capability. [Learn more about the prerequisite](#).
- Pay-as-you-go billing needs to be set up to configure policies for non-Microsoft 365 data sources. [Learn more about pay-as-you-go billing](#).

View role groups

Location	Scope	Actions
☒ Exchange email	All groups	Edit
☒ SharePoint sites	All sites	Edit
☒ OneDrive accounts	All users & groups	Edit

Back Next Cancel

Microsoft Purview

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Policy mode

Finish

Define policy settings

Decide if you want to use the default settings from the template you selected to quickly set up a policy or configure custom rules to refine your policy further.

Define policy settings

☐ Review and customize default settings from the template.

☒ Create or customize advanced DLP rules

4.2 DLP-001: Block External Sharing of Confidential Data

This policy prevents any document or email labeled Confidential or Highly Confidential from being shared with people outside the organization. The rule uses two conditions: first, 'Content is shared with people outside your organization' (establishing the sharing context), then 'Content contains sensitivity label Confidential or Highly Confidential' (identifying what to protect). When triggered, external sharing is blocked, a policy tip notifies the user explaining why, and an incident report is generated for the security team. Deployed in test mode for impact evaluation before enforcement.

Microsoft Purview | Search for solutions, users, articles, and more | Copilot

Data loss prevention > Create policy

Template or custom policy
Name
Admin units
Locations
Policy settings
Advanced DLP rules
Policy mode
Finish

Create rule

Use rules to define the type of sensitive information you want to protect. If the content matches many rules, the most restrictive one will be enforced. [Learn more about rules](#)

Name *
DLP-001: Block External Sharing of Confidential Content

Description

Conditions
Define the conditions that must be met for this policy to be applied. Include specific content, senders, and recipients that you want the rule to detect. For more complex rules, create groups to exclude or include items. [Learn how the condition builder works](#)

☒ Quick summary

Content contains

Group name *	Group operator
Default	Any of these
Sensitivity labels	
Confidential/All Employees	
Confidential/Trusted People	
Highly Confidential/All Employees	
Highly Confidential/Specified People	
Add	

AP: Create rules...

Microsoft Purview | Search for solutions, users, articles, and more | Copilot

Data loss prevention > Create policy

Template or custom policy
Name
Admin units
Locations
Policy settings
Advanced DLP rules
Policy mode
Finish

Create rule

Actions
Use actions to protect content when the conditions are met.

Restrict access or encrypt the content in Microsoft 365 locations

☒ Block users from receiving email, or accessing shared SharePoint, OneDrive, Teams files, Fabric and Power BI items.
By default, users are blocked from sending Teams chats and channel messages that contain the type of content you're protecting. But you can choose who is blocked from receiving emails or accessing files shared from SharePoint, OneDrive, Teams, Fabric and Power BI items.

☐ Block everyone

☒ Block only people outside your organization

+ Add an action

User notifications
Use notifications to inform your users and help educate them on the proper use of sensitive info.

☒ On

☒ Email notifications

Preview and edit notification email

☒ Notify the user who sent, shared, or last modified the content.

☐ Notify these people:

☒ Attach matching email message to the notification (applies only to Exchange)

☒ Policy tips

☐ Learn when policy tips appear: Although the rule will always be enforced, support and behavior for policy tips vary across apps, platforms, and user licenses. For example, policy tips might not appear to users with unsupported licenses or Outlook versions. [Learn more about when policy tips appear](#)

☐ Customize the policy tip text

Save Cancel

Microsoft Purview | Search for solutions, users, articles, and more | Copilot

Data loss prevention > Create policy

Template or custom policy
Name
Admin units
Locations
Policy settings
Advanced DLP rules
Policy mode
Finish

Create rule

Allow overrides from Microsoft 365 files and Microsoft Fabric items

☐ Allow users to override policy restrictions in Fabric (including Power BI), Exchange, SharePoint, OneDrive, and Teams.

Incident reports
Use this severity level in admin alerts and reports: Select an option

Send an alert to admins when a rule match occurs.
☒ On

Send email alerts to these people (optional)
+ Add or remove users

☒ Send alert every time an activity matches the rule

☐ Send alert when the volume of matched activities reaches a threshold

☐ Instances more than or equal to 15 matched activities

☐ Volume more than or equal to 0 MB

During the last 60 minutes

For All users

Use email incident reports to notify you when a policy match occurs.
☒ Off

Additional options

☐ If there's a match for this rule, stop processing additional DLP policies and rules.

Evaluate rule per component (Email body and each individual attachment will be considered an individual entity for rule evaluation)
☒ Off

Set the order in which this rule will be selected for evaluation
Priority: 0

Save Cancel

Microsoft Purview

Search for solutions, users, articles, and more

Copilot

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Advanced DLP rules

Policy mode

Finish

Customize advanced DLP rules

The rules here are made up of conditions and actions that define the protection requirements for this policy. You can edit existing rules or create new ones.

+ Create rule *

1 item

Name	Status			
DLP-001: Block External Sharing of Confidential Content Conditions Content contains any of these sensitivity labels: Confidential/All Employees, Confidential/Trusted People, Highly Confidential/All Employees, Highly Confidential/Specified People Evaluate predicate for Message or attachment Actions Notify users with email and policy tips Restrict access to the content for external users Send alerts to Administrator	On			

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Policy mode

Finish

Policy mode

You can test this policy before turning it on to verify if it needs improvement or if it meets all your objectives. If you turn the policy on right away, you can edit it later and safely test those changes in simulation mode.

Run the policy in simulation mode

We'll show you items that match the policy's conditions to help you evaluate its impact. Your data won't be affected; the policy stays off while in simulation mode. [Learn more about simulation mode](#)

Show policy tips while in simulation mode.

Turn the policy on if it's not edited within fifteen days of simulation

Turn the policy on immediately

After creation, the policy will be turned on and begin enforcing changes once applied to the location.

Leave the policy turned off

Decide to test or activate the policy later.

Microsoft Purview

Search for solutions, users, articles, and more

Copilot

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Policy mode

Finish

Review and finish

Create the policy if these details look fine. Otherwise, adjust the settings to better meet your needs.

Basic settings

The information to protect

Custom policy

Edit

Name

DLP-001: Block External Sharing of Confidential Content

Edit

Description

Edit

Locations

Consider adding Teams as a location to protect the accidental sharing of sensitive info in Teams messages.

Update locations

Exchange email

SharePoint sites

OneDrive accounts

Mode

Run the policy in simulation mode

Edit

Rules

DLP-001: Block External Sharing of Confidential Content

Edit

Back

Submit

Technical Note: The DLP engine requires the sharing context condition ('Content is shared with people outside your organization') as the first condition with operator 'AND' before content conditions. This ensures the policy evaluates sharing behavior before checking content sensitivity - a logical prerequisite for the 'Block External' action.

ISO 27001: A.8.12 - Data leakage prevention. SOC 2: CC6.1 - Prevent unauthorized disclosure. PCI DSS: Requirement 3, 7 - Restrict access to cardholder data.

4.3 DLP-002: Payment Card Data Protection

This policy specifically targets credit card number patterns using Microsoft's built-in PCI sensitive information type detector. When a document or email containing credit card data is shared externally, the policy blocks the

sharing, notifies the user with a policy tip, and generates an incident report. This directly supports PCI DSS compliance by preventing cardholder data from leaving the organization's control boundary.

Microsoft Purview

Search for solutions, users, articles, and more

Copilot

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Policy mode

Finish

Name your DLP policy

Create a DLP policy to detect sensitive data across locations and apply protection actions when the conditions match.

Name * ⓘ

DLP-002: Payment Card Data Protection

Description

Helps detect the presence of information subject to PCI Data Security Standard (PCI DSS), including information like credit card or debit card numbers.

Microsoft Purview

Search for solutions, users, articles, and more

Copilot

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Policy mode

Finish

Choose where to apply the policy

We'll apply the policy to data that's stored in the locations you choose.

ⓘ The Instances policy location will be retired in early 2027 as part of the Microsoft Defender for Cloud Apps file policy retirement. For non-Microsoft connected apps, create new DLP policies in Purview using a specific location (for example, Box or Google Workspace) instead.

ⓘ Before you continue creating this Microsoft Purview DLP policy, turn off or delete any Microsoft Defender for Cloud Apps file policies for non-Microsoft locations. Running both at once can cause unexpected policy enforcement.

ⓘ Conditional Access policy not found. Conditional access policies are required for managed app policies. [Learn more.](#) Check your conditional access policy settings or create a new one in Entra ID [here](#).

ⓘ Edge for business protection settings not found. This is required for managed app policies. [Learn more.](#) Check your Edge for business settings [here](#).

ⓘ If your role group permissions are restricted to a specific set of users or groups, you'll only be able to apply this policy to those users or groups. [Learn more about role group permissions.](#)

View role groups

ⓘ Protecting sensitive info in on-premises repositories (SharePoint sites and file shares) is now in preview. Note that there are prerequisite steps needed to support this new capability. [Learn more about the prerequisites](#)

ⓘ Pay-as-you-go billing needs to be set up to configure policies for non-Microsoft 365 data sources. [Learn more about pay-as-you-go billing](#)

Location	Scope	Actions
☒ Exchange email	All groups	Edit
☒ SharePoint sites	All sites	Edit
☒ OneDrive accounts	All users & groups	Edit

Back

Next

Cancel

Microsoft Purview

Search for solutions, users, articles, and more

Copilot

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Policy mode

Finish

Define policy settings

Decide if you want to use the default settings from the template you selected to quickly set up a policy or configure custom rules to refine your policy further.

Define policy settings

☒ Review and customize default settings from the template. ⓘ

Credit Card Number

☐ Create or customize advanced DLP rules ⓘ

Microsoft Purview

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Info to protect

Protection actions

Customize access and override settings

Policy mode

Finish

Info to protect

This policy will protect content that matches these conditions. Review them and make any necessary changes. For example, you can edit the conditions to detect additional sensitive info or content that has specific sensitivity or retention labels applied.

Content contains any of these sensitive info types:
Credit Card Number

[Edit](#)

☒ Detect when this content is shared from Microsoft 365: ⓘ

☒ With people outside my organization

☐ Only with people inside my organization

Microsoft Purview

Data loss prevention > Create policy

Template or custom policy

Name

Admin units

Locations

Policy settings

Policy mode

Finish

Review and finish

Create the policy if these details look fine. Otherwise, adjust the settings to better meet your needs.

Basic settings

The information to protect
PCI Data Security Standard (PCI DSS) [Edit](#)

Name
DLP-002: Payment Card Data Protection [Edit](#)

Description [Edit](#)

Locations [Edit](#)

ⓘ Consider adding Teams as a location to protect the accidental sharing of sensitive info in Teams messages. [Update locations](#)

Exchange email
SharePoint sites
OneDrive accounts

Mode
Run the policy in simulation mode [Edit](#)

Rules
Low volume of content detected DLP-002: Payment Card Data Protec
High volume of content detected DLP-002: Payment Card Data Prote [Edit](#)

[Back](#) [Submit](#)

4.4 All DLP Policies - Summary View

Both DLP policies deployed in test mode, ready for impact evaluation before enforcement.

Enforce policy

+ Create policy | Export policies | Refresh | Copilot

5 items | Search | Customize columns

Name	Priority	Mode	Policy sync status	Last modified
Default Office 365 DLP policy	0	On	Sync completed	Sep 14, 2026 3:02 PM
Default DLP policy - Protect sensitive M365 Copilot interactions	1	In simulation with notifications	Sync completed	Sep 14, 2026 3:02 PM
Default policy for Teams	2	On	Sync completed	Sep 14, 2026 3:02 PM
DLP-001: Block External Sharing of Confidential Content	3	In simulation with notifications	Sync completed	Sep 20, 2026 3:27 PM
DLP-002: Payment Card Data Protection	4	In simulation with notifications	Sync in progress	Sep 20, 2026 6:40 PM

PCI DSS: Requirement 3 - Protect stored cardholder data. Requirement 4 - Encrypt transmission of cardholder data. HIPAA: Prevent unauthorized PHI disclosure.

5. Cross-Project Integration - Identity + Endpoint + Data Protection

This project completes the third layer of a unified security architecture spanning identity (Project 1 - Entra ID), endpoint (Project 2 - Intune), and data (Project 3 - Purview). Together, these three projects demonstrate the zero-trust data protection chain:

Layer 1 - Identity (Entra ID): WHO is accessing the data? Authenticated user with MFA, part of the correct security group, no risky sign-in signals.

Layer 2 - Endpoint (Intune): WHAT device are they using? Compliant device with encryption, up-to-date OS, active antivirus, meeting the organization's security baseline.

Layer 3 - Data (Purview): WHAT data are they accessing, and what can they do with it? Classified by sensitivity level, marked with headers and watermarks, protected by DLP policies preventing unauthorized external sharing.

This three-layer approach ensures that even if an attacker compromises one layer (e.g., steals credentials), the other layers still protect the organization's data. A stolen password without a compliant device fails the Conditional Access policy. A compromised device without the right identity fails authentication. And even with both identity and device access, DLP policies prevent sensitive data from being exfiltrated. This is defense in depth operationalized through the Microsoft security stack.

6. Conclusion

This project demonstrates end-to-end data classification and data loss prevention implementation in Microsoft Purview - from a four-tier sensitivity label taxonomy through automated sensitive data detection to enforcement-level DLP policies protecting against unauthorized data disclosure. Every configuration maps to specific compliance requirements across ISO 27001, SOC 2, PCI DSS, and HIPAA.

Combined with the IAM Framework (Project 1) and Endpoint Compliance Program (Project 2), these three projects establish a complete zero-trust security architecture covering identity, endpoint, and data protection layers - demonstrating both the technical implementation capability and the compliance-mapping discipline required for enterprise GRC roles.

This is Project 3 of 5 in the Microsoft Security Portfolio. Subsequent projects cover Security Posture Assessment (Microsoft Defender + Nessus) and Incident Response Playbooks.